UGANDA NATIONAL EXAMINATIONS BOARD

PRIMARY LEAVING EXAMINATION

2010

MATHEMATICS

Time Allowed: 2 hours 30 minutes

Random No.						Personal No.		

Candidate's Name: .....

Candidate's Signature: .....

District ID No. | | | | |

Read the following instructions carefully:

1. Do not write your **school** or **district name** anywhere on this paper.
2. This paper has two Sections: **A** and **B**. Section **A** has **30** questions and Section **B** has **12** questions. The paper has **15 printed** pages.
3. Answer **all** questions. **All** working for both Sections **A** and **B** must be shown in the spaces provided.
4. All working **must** be done using a **blue** or **black** ball point pen or ink. Any work done in pencil other than graphs and diagrams will **not** be marked.
5. No **Calculators** are allowed in the examination room.
6. Unnecessary **changes** in your work and handwriting that cannot be easily read may lead to **loss of marks**.
7. Do not fill anything in the table indicated **"FOR EXAMINERS' USE ONLY"** and in the boxes inside the question paper.

FOR EXAMINERS' USE ONLY		
QN. NO.	**MARKS**	**EXR'S NO.**
1 – 10		
11 – 20		
21 – 30		
31 – 32		
33 – 34		
35 – 36		
37 – 38		
39 – 40		
41 – 42		
TOTAL		

©UNEB MTC

© 2010 prepared by Tr. Micah: 0760886093

Turn Over

SECTION A: 40 MARKS

Answer **all** *questions in this section*

Questions ***1*** *to* ***20*** *carry two marks each.*

1. Workout: $10 \div 2$

2. Simplify: $2x + 3x$

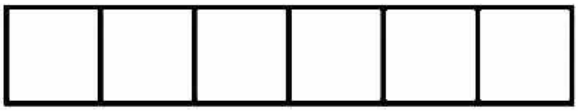
3. Write in figures: Sixty-one thousand.

4. Given set A = {a, b, f, k} and set B = {a, c, k}, find n (A∪B)

5. Simplify: $-5 + \bar{}2$

6. Write 49 in Roman numerals.

7. Shade $\frac{1}{2}$ of the drawing below.



8. Using a pair of compasses, a ruler and a pencil only, bisect the line below.



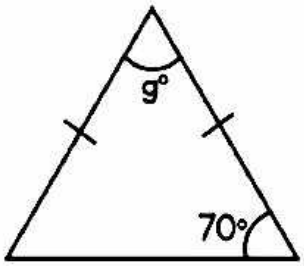
9. What is the value of 5 in the figure 65011?

10. Change 2.5 metres to centimetres.

11. Cards labelled 1 to 5 are folded, put in a bucket and mixed up. What is the probability of picking a card having a prime number?

12. Seven children had the following ages: 7, 3, 6, 2, 5, 1 and 4. Find their mean age.

13. In the triangle below, find the size of angle *g* in degrees.



14. Workout:

$$\begin{array}{r} 165 \\ \times \quad 4 \\ \hline \\ \hline \end{array}$$

15. Given that a = -3 and b = 4, find the value of 2a + 2b

16. Find the next number in the sequence:

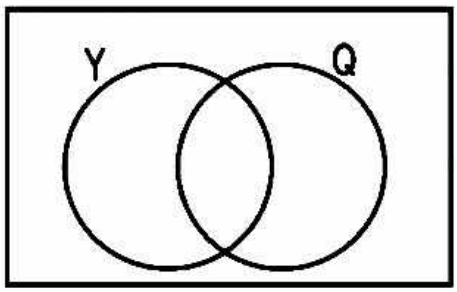
23, 19, 16, 14, ...

17. A fifty-minute test started at 9:50am. At what time did it end?

18. Solve: $4p - 4 = 20$

19. In a line of vehicles, a bus was the 7th from each end of the line. How many vehicles were in the line?

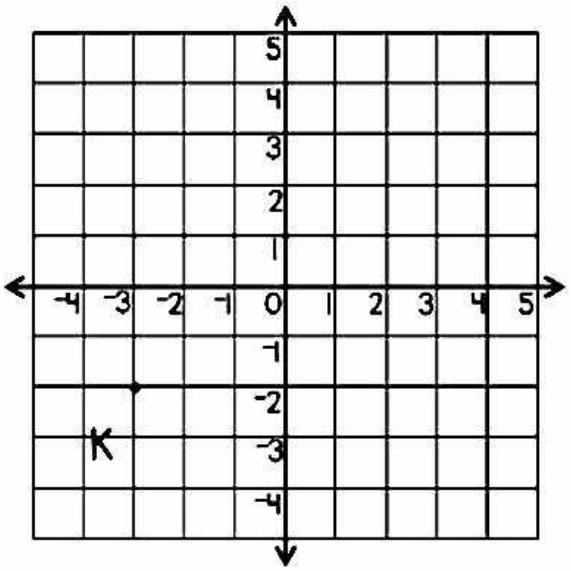
20. In the Venn diagram below, shade the area (YUQ)



21. Workout: $\frac{5}{12} - \frac{5}{9}$

22. Change 11_{ten} to base two.

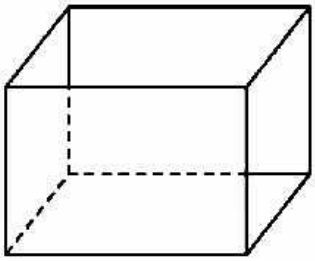
23. In the diagram below, find the co-ordinates of point K.



24. A boy ran a distance of 6km in 45 minutes. What was his speed in kilometres per hour?

25. Arrange the following decimals in order beginning from the smallest: 0.11, 0.5 and 0.03

26. How many edges does the figure below have?

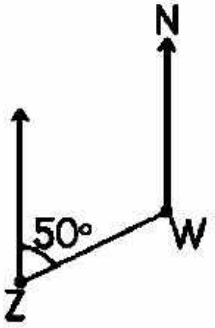


27. Workout: 6702 — 4865

28. Four packets of mango juice cost sh. 2,000. What is the cost of seven similar packets?

29. In a P.7 class, $\frac{2}{5}$ of the pupils are girls. If there are 150 pupils in class, find the number of boys.

30. In the diagram below, find the bearing of town Z from town W.

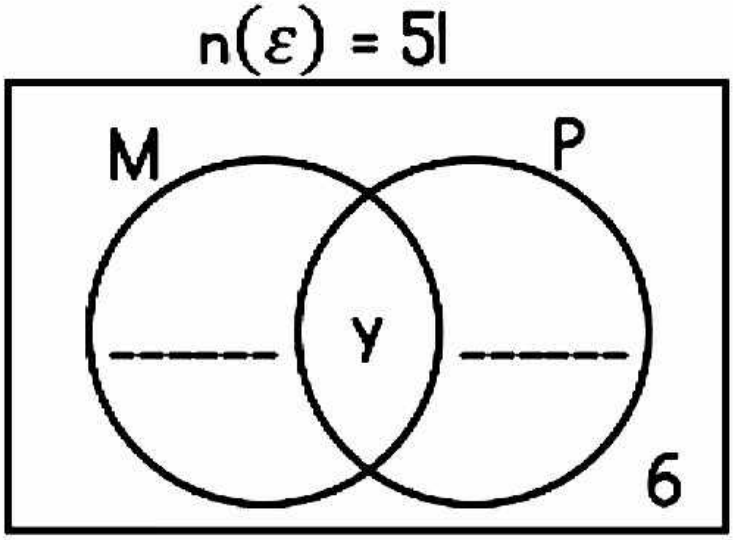


SECTION B: 60 MARKS

Answer **all** *questions in this section*
Marks for each question are indicated in brackets

31. In a class party of 5l pupils, 28 drank mirinda (M), 29 drank pepsi (P), *y* drank both mirinda and pepsi while 6 did not drink any of the two sodas.

(a) Use the information given above to complete the Venn diagram below.



(b) Find the number of pupils who drank one type of soda only.

32. (a) Using a ruler, a pencil and a pair of compasses only, construct a rectangle ABCD in which AB = 8cm and BC = 7cm.

(b) Measure the length of diagonal AC.

(c) Measure the angle BAC.

33. The table below shows the arrival and departure time of a bus that travels from Kampala to Hoima daily.

Town	**Arrival time**	**Departure time**
Kampala		7:30 a.m.
Busunju	8:10 a.m.	8:30 a.m.
Bukomero	9:30 a.m.	9:45 a.m.
Kiboga	10:15 a.m.	10:40 a.m.
Hoima	11:40 a.m.	

(a) At what time does the bus leave Kampala?

(b) How long does the bus stay at Bukomero?

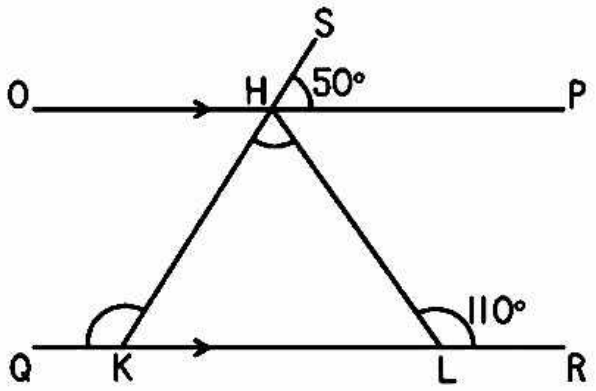
(c) How long does the bus take to travel from Bukomero to Kiboga?

(d) Find the total time taken by the bus to travel from Kampala to Hoima?

34. (a) Solve: $2m + 3 = 18 - m$

(b) Solve: $2(3x - 1) - 4(x - 1) = 4$

35. In the diagram below, OP is parallel to QR. HKL is a triangle, angle HLR = 110° and angle SHP = 50°. Study it and answer questions that follow.



Find the size of:

(a) angle *y*.

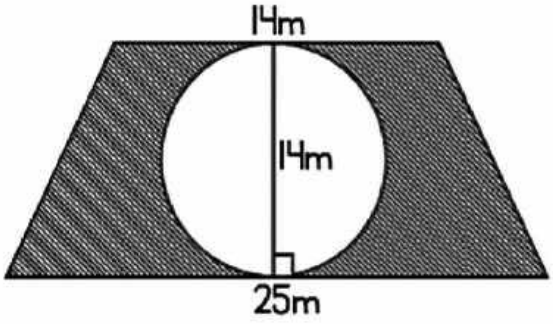
(b) angle *m*.

36. (a) Find the number which has been expanded below.

(b) Change 1011_{two} to base ten.

(c) Find the value of x: 3 + 3 = x (finite 4)

37. Find the area of the shaded part in the diagram below. (use $\pi = \frac{22}{7}$)



38. The mean of the scores 8, 9, 6, 4 and x is 6.

(a) Find the value of x.

(b) What is the median score?

(c) Find the probability that a score picked at random is below the mean.

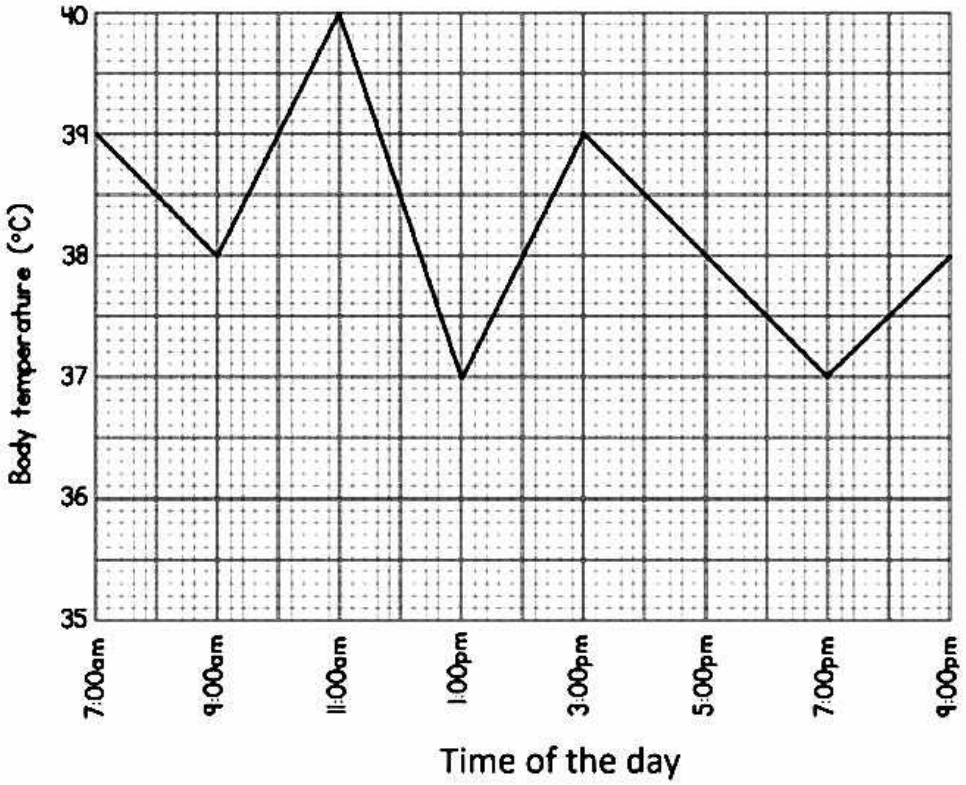
39. Makeba's car uses 8 litres of petrol for every 50km.

(a) How much petrol does it need for a journey of 325km?

(b) If one litre of petrol costs sh. 2,900, how much money will he spend on petrol needed to run the car for $\frac{1}{2}$ hours at a speed of 50km per hour?

40. The graph below shows the changes in body temperature of a patient in a hospital recorded after every two hours in a day. Use it to answer questions that follow.

Time of the day



(a) That was the highest temperature recorded?

(b) Find the range in the recorded body temperature.

(c) Workout the average body temperature of the patient from 3:00pm to 9:00pm.

41. A man spends $\frac{1}{3}$ of his salary on food, $\frac{1}{9}$ on clothing, $\frac{1}{6}$ on medical, $\frac{1}{18}$ on house rent and banks the rest which is sh 35,000.

(a) What fraction of his salary does he bank?

(b) How much money does he earn as salary?

42. Simplify:

(a) $n^2 \times n$

(b) $m^6 \div m^3$

(c) $\frac{a^2 \times a^5}{a^3}$